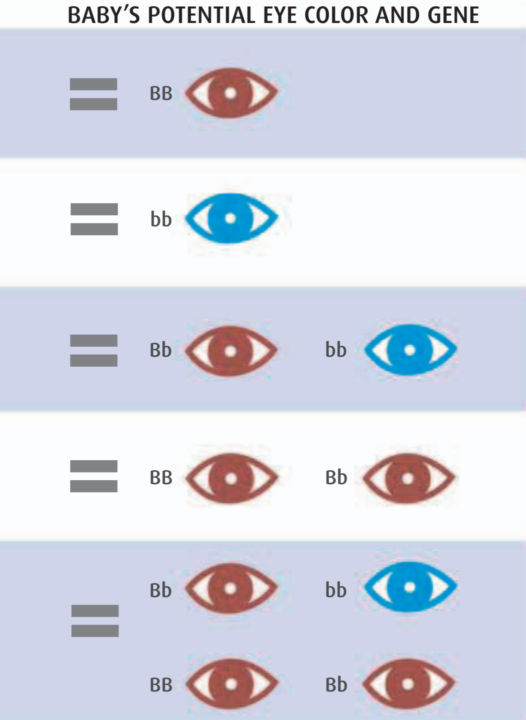
is replicated for every possible trait, including hair color, skin tone, dimples, and so on.

BABY'S POTENTIAL EYE COLOR AND GENE



Q My baby's father is bald. Does this mean that if I have a boy, he will go bald, too?

The genetic inheritance of baldness is still not fully understood. For a long time it was thought that male pattern baldness (baldness that runs in families and begins with a receding hairline) was inherited from the mother's side. So, a baby boy was assumed to inherit baldness via his mother from his maternal grandfather rather than from his own father. However, anecdotal evidence shows that this isn't always the case, which has led researchers to look for a pattern that confirms heredity directly from father to son. Although there doesn't yet seem to be evidence that the faulty gene lies within the Y chromosome, there is some evidence that mutations in genes on chromosome 20 (which can come from a baby's mother or father) may increase the risk of a boy inheriting baldness. Environmental factors (including diet and stress levels) play a role, too. In short, while the percentages suggest that boys born to bald fathers are likely to go bald themselves, there are no certainties. As far as girls going bald is concerned, it's thought that high levels of female hormones make sure the baldness gene stays inactive.

Male pattern baldness This is the most common type of hair loss in men. It is generally thought to be caused by hormones and a genetic predisposition.

Q What causes albinism?

This is a rare genetic condition, where the genes that control melanin production are faulty.

As a result the proteins in the genes don't get the message to trigger the release of melanin in the hair, eyes, and skin, which leads to lack of pigmentation. Those affected tend to be very pale in color, have pink or pale gray eyes, and white hair. It can cause sight problems and sensitivity to light, because eyes need melanin to make a healthy retina (the "screen" at the back of the eye). Albinism is inherited through autosomal recessive inheritance (in which both parents carry the faulty gene and the baby inherits it from both sides) and X-linked inheritance (in which the faulty gene is attached to the X chromosome in one parent).

X-LINKED INHERITANCE

An X-linked inheritance refers to conditions that occur as a result of a faulty gene within the X chromosome. The box below explains how this works.

MOTHER (XX)	FATHER (XY)
A baby girl has a 50-percent chance of becoming a carrier of the faulty gene (and a 50-percent chance of not inheriting a faulty gene at all), but would not herself have the condition. This is because girls have two X chromosomes (XX)—in this case, one healthy and one carrying the faulty gene. The healthy X chromosome compensates for the carrier, and the baby girl appears healthy.	A baby girl will definitely become a carrier, because in order to become a girl (XX) she must inherit the damaged X chromosome from her father.
A baby boy has one X chromosome and one Y chromosome (XY), inherited from his mother and father, respectively, giving him a 50-percent chance that he will inherit the X chromosome from his mother that carries the faulty gene. If this happens he will develop albinism, because he has no healthy X chromosome to compensate.	A baby boy does not become a carrier, because to become a boy he must inherit the Y chromosome from his father.